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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

Discrete Fourier Transform and Fast Fourier Transform

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

18	15	2008 – 2024	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	4 questions
Marks 2	14 questions
Answer key	at the end

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Marks 1 4 questions

Q1. GATE ECE 2023

Let $w^4 = 16j$. Which of the following cannot be a value of w?

(A) $2e^{\frac{j2\pi}{8}}$ (B) $2e^{\frac{j\pi}{8}}$ (C) $2e^{\frac{j5\pi}{8}}$ (D) $2e^{\frac{j9\pi}{8}}$

Open solution

Q2. GATE ECE 2022

The Fourier transform X(jω) of the signal $x(t) = \frac{t}{(1+t^2)^2}$ is _____.

(A) $\frac{\pi}{2j}we^{-|\omega|}$ (B) $\frac{\pi}{2}we^{-|\omega|}$

(C) $\frac{\pi}{2j}e^{-|\omega|}$ (D) $\frac{\pi}{2}e^{-|\omega|}$

Open solution

Q3. GATE ECE 2021

Consider two 16-point sequences x[n] and h[n]. Let the linear convolution of x[n] and h[n] be denoted by y[n], while z[n] denotes the 16-point inverse discrete Fourier transform (IDFT) of the product of the 16-point DFTs of x[n] and h[n]. The value(s) of k for which z[k] = y[k] is/are :

(A) k = 0, 1, 2, ..., 15 (B) k = 0 and k = 15

(C) k = 0 (D) k = 15

Open solution

Q4. GATE ECE 2010

For an N-point FFT algorithm with N = 2^m which one of the following statements is TRUE?

(A) It is not possible to construct a signal flow graph with both input and output in normal order.

(B) The number of butterflies in the mth stage is N/m.

(C) In-place computation requires storage of only 2N node data.

(D) Computation of a butterfly requires only one complex multiplication.

Open solution

Marks 2 14 questions

Q5. GATE ECE 2024

The relationship between any N-length sequence $x[n]$ and its corresponding N-point discrete Fourier transform $X[k]$ is defined as

$X[k] = \mathcal{F}\{x[n]\}$.

Another sequence $y[n]$ is formed as below

$y[n] = \mathcal{F}\{\mathcal{F}\{\mathcal{F}\{\mathcal{F}\{x[n]\}\}\}\}$.

For the sequence $x[n] = \{1, 2, 1, 3\}$, the value of $Y[0]$ is _____.

Open solution

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Q6. GATE ECE 2023

Let an input $x[n]$ having discrete time Fourier transform $x(e^{j\Omega}) = 1 - e^{-j\Omega} + 2e^{-3j\Omega}$ be passed through an LTI system. The frequency response of the LTI system is $H(e^{j\Omega}) = 1 - \frac{1}{2}e^{-j2\Omega}$. The output $y[n]$ of the system is

Open solution

(A) $\delta[n] + \delta[n-1] - \frac{1}{2}\delta[n-2] - \frac{5}{2}\delta[n-3] + \delta[n-5]$

(B) $\delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] - \frac{5}{2}\delta[n-3] + \delta[n-5]$

(C) $\delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]$

(D) $\delta[n] + \delta[n-1] + \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] + \delta[n-5]$

Q7. GATE ECE 2022

Open solution

For a vector $\overline{x}$ = [x[0], x[1], ..., x[7]], the 8-point discrete Fourier transform (DFT) is denoted by $\overline{X}$ = DFT($\overline{x}$) = [X[0], X[1], ..., X[7]], where

$$X[k] = \sum_{n=0}^{7} x[n] \exp\left(-j\frac{2\pi}{8}nk\right).$$

Here, $j = \sqrt{-1}$. If $\overline{x}$ = [1, 0, 0, 0, 2, 0, 0, 0] and $\overline{y}$ = DFT (DFT($\overline{x}$)), then the value of y[0] is _____ (rounded off to one decimal place).

Q8. GATE ECE 2020

Open solution

A finite duration discrete-time signal $x[n]$ is obtained by sampling a continuous - time signal $x(t) = \cos(200\pi t)$ at sampling instants $t = \frac{n}{400}, n = 0, 1, \ldots, 7$. The 8-point discrete Fourier transform (DFT) is defined as

$X[k] = \sum_{n=0}^{7} x[n]e^{-j\pi nk/4}$ for $k = 0, 1, \ldots, 7$

Which one of the following statements is TRUE?

(A) Only $X(3)$ and $X(5)$ are non-zero.

(B) Only $X(4)$ is non-zero.

(C) All $X(K)$ are non-zero.

(D) Only $X(2)$ and $X(6)$ are non-zero.

Q9. GATE ECE 2016 Set 2

Open solution

The Discrete Fourier Transform (DFT) of the 4-point sequence
$x\,[n]$= {x[0], x[1], x[2], x[3]}
= {3, 2, 3, 4 } is
x[k] = {X[0], X[1], X[2], X[3]}
= {12, 2j, 0, -2j }
If X_1 [k] is the DFT of the 12- point sequenceX_1[n] = {3, 0, 0, 2, 0, 0, 3, 0, 0, 4, 0, 0 },
The value of $\left|\frac{X_1[8]}{X_1[11]}\right|$ is----------.

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Q10. GATE ECE 2016 Set 3

Open solution

A continuous-time speech signal $x_a(t)$ is sampled at a rate of 8 kHz and the samples are subsequently grouped in blocks, each of size N. The DFT of each block is to be computed in real time using the radix-2 decimation-in-frequency FFT algorithm. If the processor performs all operations sequentially, and takes 20 μs for computing each complex multiplication (including multiplications by 1 and −1) and the time required for addition/ subtraction is negligible, then the maximum value of N is _____.

Q11. GATE ECE 2015 Set 1

Open solution

Two sequences [a, b, c] and [A, B, C] are related as,

$$\begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^{-1} & W_3^{-2} \\ 1 & W_3^{-2} & W_3^{-4} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$ Where

$W_3 = e^{j\frac{2\pi}{3}}$.

if another sequence $[p,\ q,\ r]$ is derived as,

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^1 & W_3^2 \\ 1 & W_3^2 & W_3^4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & W_3^2 & 0 \\ 0 & 0 & W_3^4 \end{bmatrix} \begin{bmatrix} A/3 \\ B/3 \\ C/3 \end{bmatrix},$$

Then the relationship between the sequences $[p,\ q,\ r]$ and $[a,\ b,\ c]$ is

(A) $[p,\ q,\ r] = [b,\ a,\ c]$

(B) $[p,\ q,\ r] = [b,\ c,\ a]$

(C) $[p,\ q,\ r] = [c,\ a,\ b]$

(D) $[p,\ q,\ r] = [c,\ b,\ a]$

Q12. GATE ECE 2015 Set 1

Open solution

Consider two real sequences with time-origin marked by the bold value,

$x_1[n] = \{1,\ 2,\ 3,\ 0\}$, $x_2[n] = \{1,\ 3,\ 2,\ 1\}$ Let $X_1(k)$ and $X_2(k)$ be 4-point DFTs of $x_1[n]$ and $x_2[n]$, respectively. Another sequence $x_3[n]$ is derived by taking 4-point inverse DFT of $X_3(k) = X_1(k)$ $X_2(k)$. The value of $x_3[2]$ is _____.

Q13. GATE ECE 2014 Set 1

Open solution

Consider a discrete time periodic signal $x[n] = \sin\left(\frac{\pi n}{5}\right)$. Let a_k be the complex Fourier serier coefficients of $x[n]$. The coefficients $\{a_k\}$ are non- zero when k = Bm $\pm$ 1, where m is any integer. The value of B is _____.

Q14. GATE ECE 2014 Set 4

Open solution

The N-point DFT X of a sequence x[n] 0 ≤ n ≤ N − 1 is given by

$$X[k] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk},\ 0 \le k \le N-1$$

Denote this relation as X = DFT(x). For N= 4 which one of the following sequences satisfies DFT (DFT(x)) = _____.

(A) x = $[1\,2\,3\,4]$

(B) x = $[1\,2\,3\,2]$

(C) x = $[1\,3\,2\,2]$

(D) x = $[1\,2\,2\,3]$

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Q15. GATE ECE 2013

Open solution

The DFT of a vector [a b c d] is the vector [α β γ δ]. Consider the product

$$[p \quad q \quad r \quad s]=[a \quad b \quad c \quad d]\begin{bmatrix} a & b & c & d \\ d & a & b & c \\ c & d & a & b \\ b & c & d & a \end{bmatrix}.$$

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The DFT of the vector [p q r s] is a scaled version of

(A) $[\alpha^2\beta^2\Upsilon^2\delta^2]$

(B) $\sqrt{\alpha}\ \sqrt{\beta}\sqrt{\gamma}\sqrt{\delta}$

(C) $[\alpha+\beta\beta+\delta\ +\gamma\gamma\ +\alpha]$

(D) $[\delta\beta\gamma\delta]$

Q16. GATE ECE 2011

Open solution

The first six points of the 8-point DFT of a real valued sequence are 5, 1 - j3, 0, 3- j4, 0 and 3+ j4. The last two points of the DFT are respectively

(A) 0, 1- j3 (B) 0, 1+ j3 (C) 1+j3, 5 (D) 1 – j3, 5

Q17. GATE ECE 2009

Open solution

The 4-point Discrete Fourier Transform (DFT) of a discrete time sequence $\{1,\ 0,\ 2,\ 3\}$ is

(A) $[0,\ -2+2j,\ 2,\ -2-2j]$

(B) $[2,\ 2+2j,\ 6,\ 2\ -2j]$

(C) $[6,\ 1-3j,\ 2,\ 1+3j]$

(D) $[6,\ -1+3j,\ 0,\ -1\ -3j]$

Q18. GATE ECE 2008

Open solution

{x(n)} is a real-valued periodic sequence with a period N. x(n) and X(k) form N-point Discrete Fourier Transform (DFT) pairs. The DFT Y(k) of the sequence

y (n) = $\frac{1}{N}\sum_{r=0}^{N-1} x\ (r)x\ (n+r\ )$ is

(A) $|X(k)|^2$

(B) $\frac{1}{N}\sum_{r=0}^{N-1} X\ (r)X^*\ (k+r\ )$

(C) $\frac{1}{N}\sum_{r=0}^{N-1} X\ (r\ )X(k+r)$

(D) 0

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Answer Key

Marks 1

1. **A**	2. **A**	3. **D**	4. **A**

Marks 2

5. **112**	6. **C**	7. **8**	8. **D**	9. **6**
10. **4096**	11. **C**	12. **11**	13. **10**	14. **B**
15. **A**	16. **B**	17. **D**	18. **A**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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